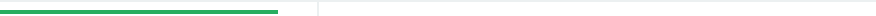




Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix



SOURCE

Our World in Data

AUDIENCE

Policy Analysts & Energy Strategists

KEY INSIGHT

The case for rapid energy transition is compelling and time-sensitive

75%

of global GHG emissions
from fossil fuels

5M+

premature deaths annually
from air pollution

Why This Matters

**Fossil Fuel Dominance**

Energy systems dominated by fossil fuels since the Industrial Revolution

**Climate Impact**

Three-quarters of global GHG emissions from burning fossil fuels for energy

**Health Crisis**

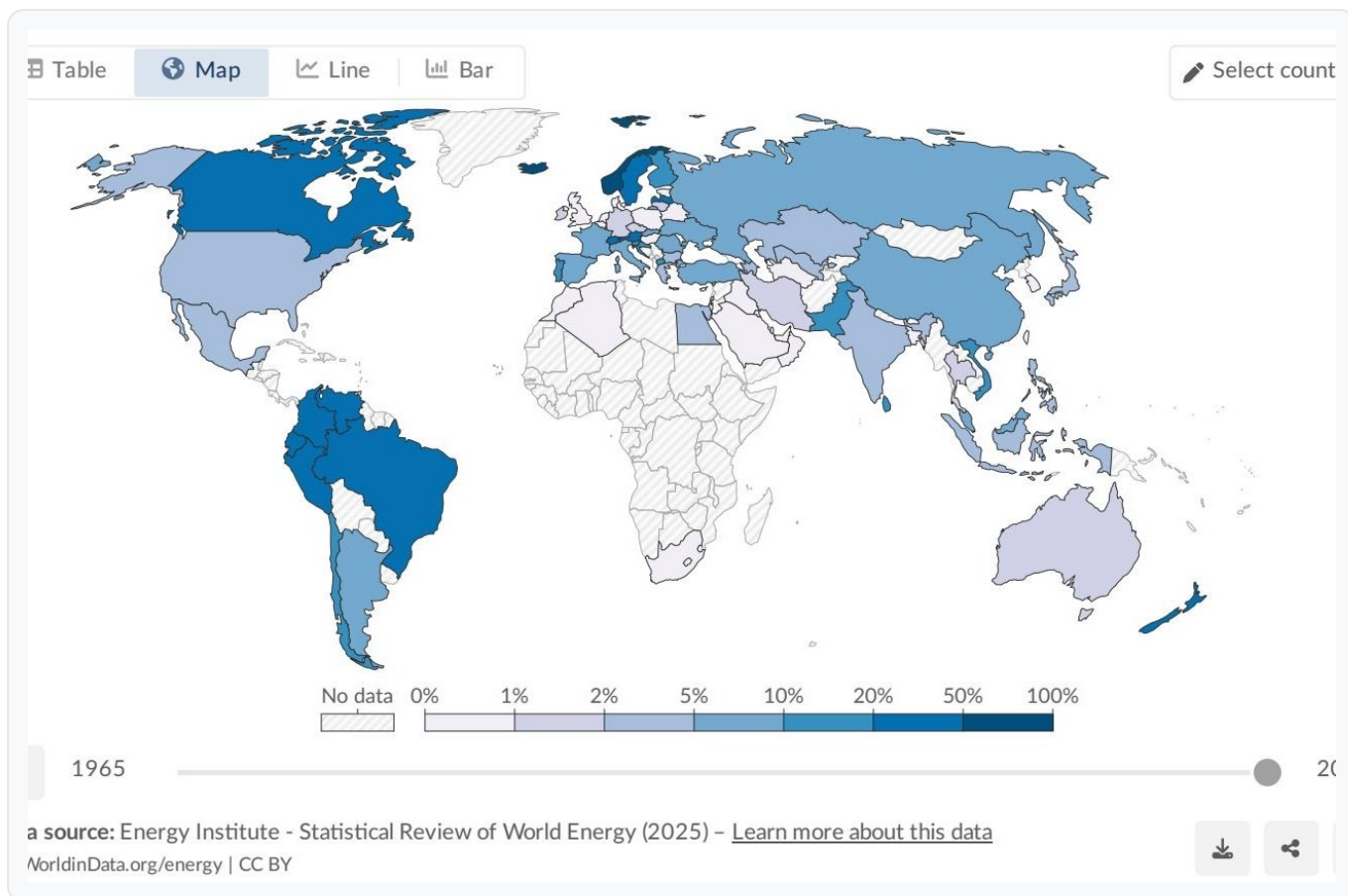
Fossil fuel combustion causes at least 5 million premature deaths per year

The Solution

Rapid shift toward low-carbon energy sources — nuclear and renewable technologies

KEY INSIGHT

Approximately one-seventh of primary energy comes from renewable technologies (2024)



What is Primary Energy?

Primary energy = electricity + transport + heating combined

Regional Leaders

Norway

High hydro capacity

Brazil

Hydropower + bioenergy

New Zealand

Hydro + geothermal

Fossil-Dependent Regions

Middle East & Parts of Asia

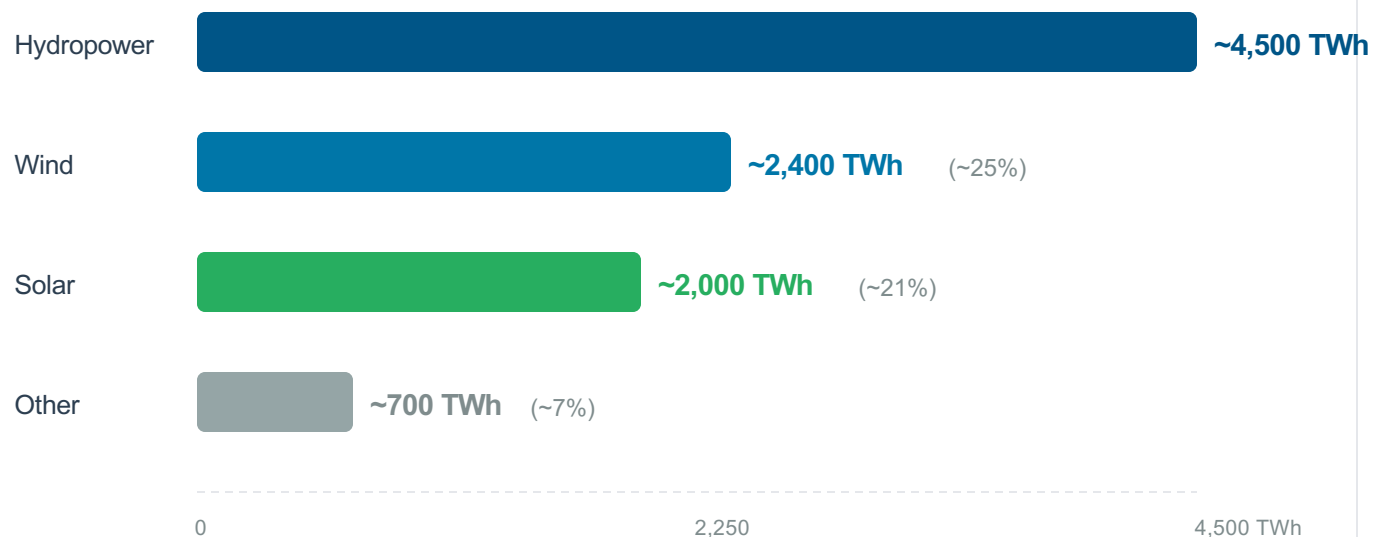
Many countries below 5% renewable share
 Heavy reliance on oil, gas, and coal

KEY INSIGHT

Wind and solar combined now nearly match hydropower output — and are growing much faster

Renewable Electricity Generation by Technology (2024)

Approximate global generation in TWh

**KEY FINDING**

Wind + Solar
≈ Hydro Output

Combined generation now nearly matches hydropower

GROWTH TREND

Acceleration Since ~2010

Solar in particular has grown dramatically
Driven by falling costs & policy support

TOTAL RENEWABLE

~9,600 TWh

Total renewable electricity generation (2024)

Hydro + Wind + Solar + Other

KEY INSIGHT

Renewables have made most progress in electricity — substantially higher than primary energy share

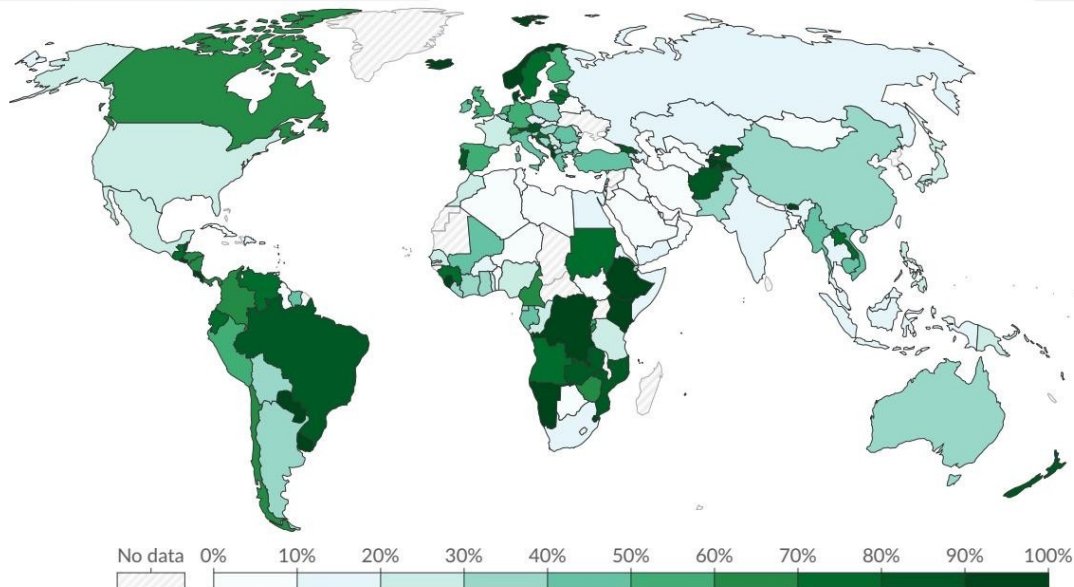
Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

Table Map Line Bar

Select countries



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) - [Learn more about this data](#)

OurWorldinData.org/energy | CC BY

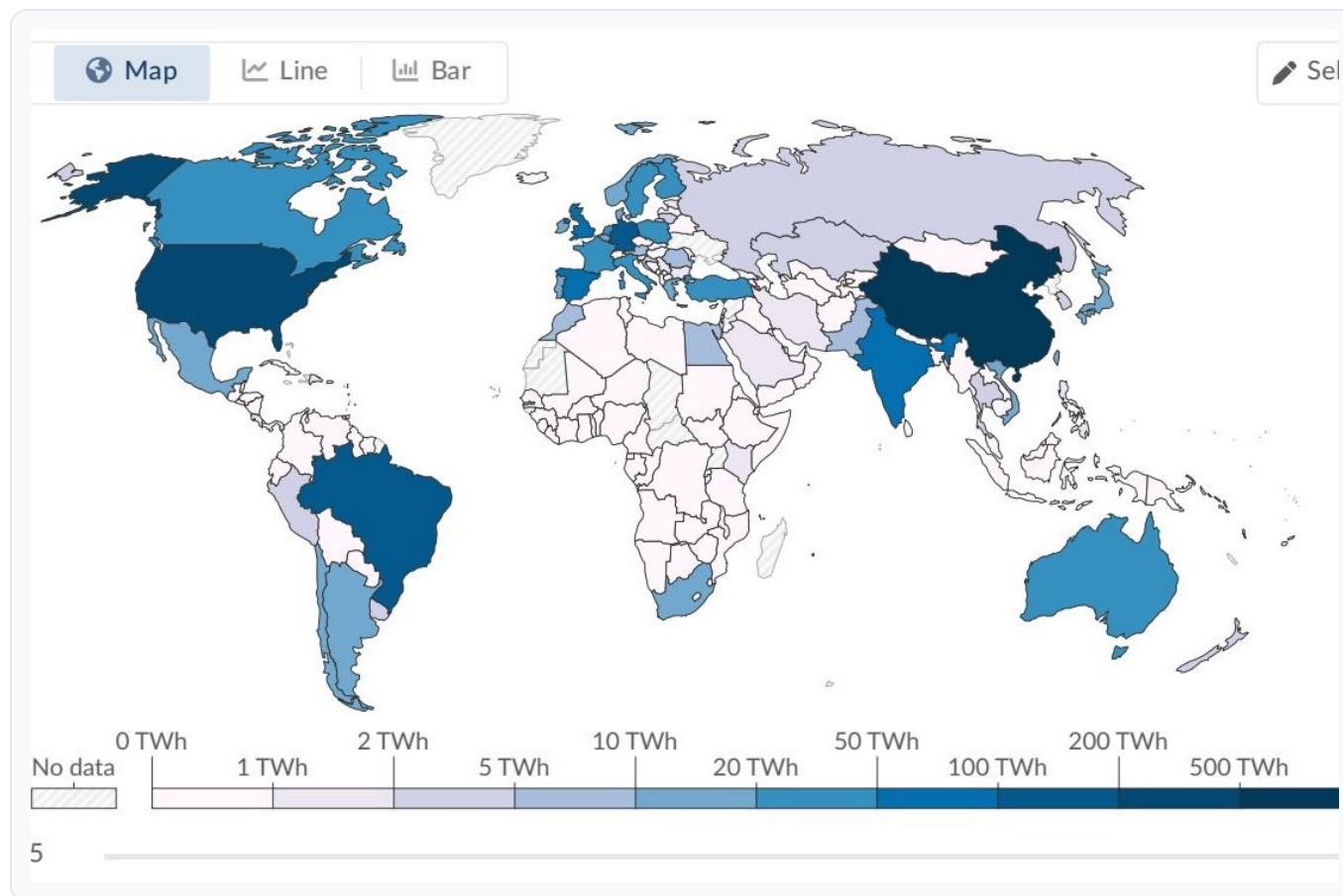
Renewable Electricity Share by Country/Region

2025 data (approximate)

Country/Region	Renewable Share	Status
Norway	~98%	Leader
Brazil	~85%	Leader
Canada	~65%	Strong
China	~35%	Growing
India	~25%	Growing
World Average	~30%	Avg
Saudi Arabia	<2%	Low

KEY INSIGHT

Hydropower accounts for roughly half of all renewable electricity generation



~50%

of renewable electricity

Top Hydropower Producers

China

World's largest

Brazil

Large capacity

Canada

Hydro-rich

Key Characteristics



Mature Technology

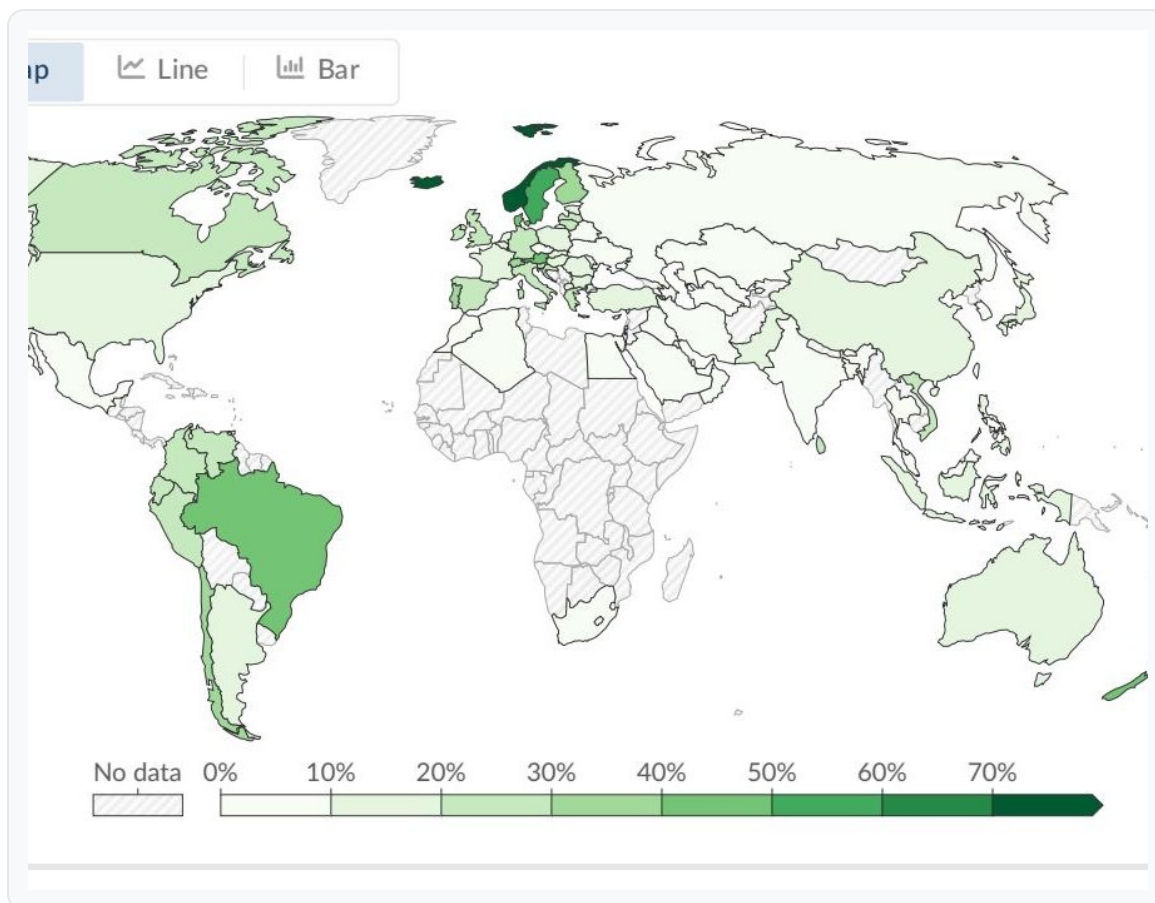
Generating electricity at scale for over a century

Growth Outlook

Growth potential limited compared to wind and solar

KEY INSIGHT

Solar is the fastest-growing source — dramatic growth since 2010 driven by falling costs and policy support

**Wind vs Solar Comparison**

Global renewable electricity generation (2024)

WIND

~2,400 TWh

~25% of renewable electricity

Strong in Europe, China, US

SOLAR

~2,000 TWh

~21% of renewable electricity

Fastest-growing globally

KEY FINDING

Wind + Solar Combined ≈ Hydro Output

Combined output nearly matches hydropower

Growth Drivers

Falling Costs

Module prices dropped



Policy Support

Subsidies & targets

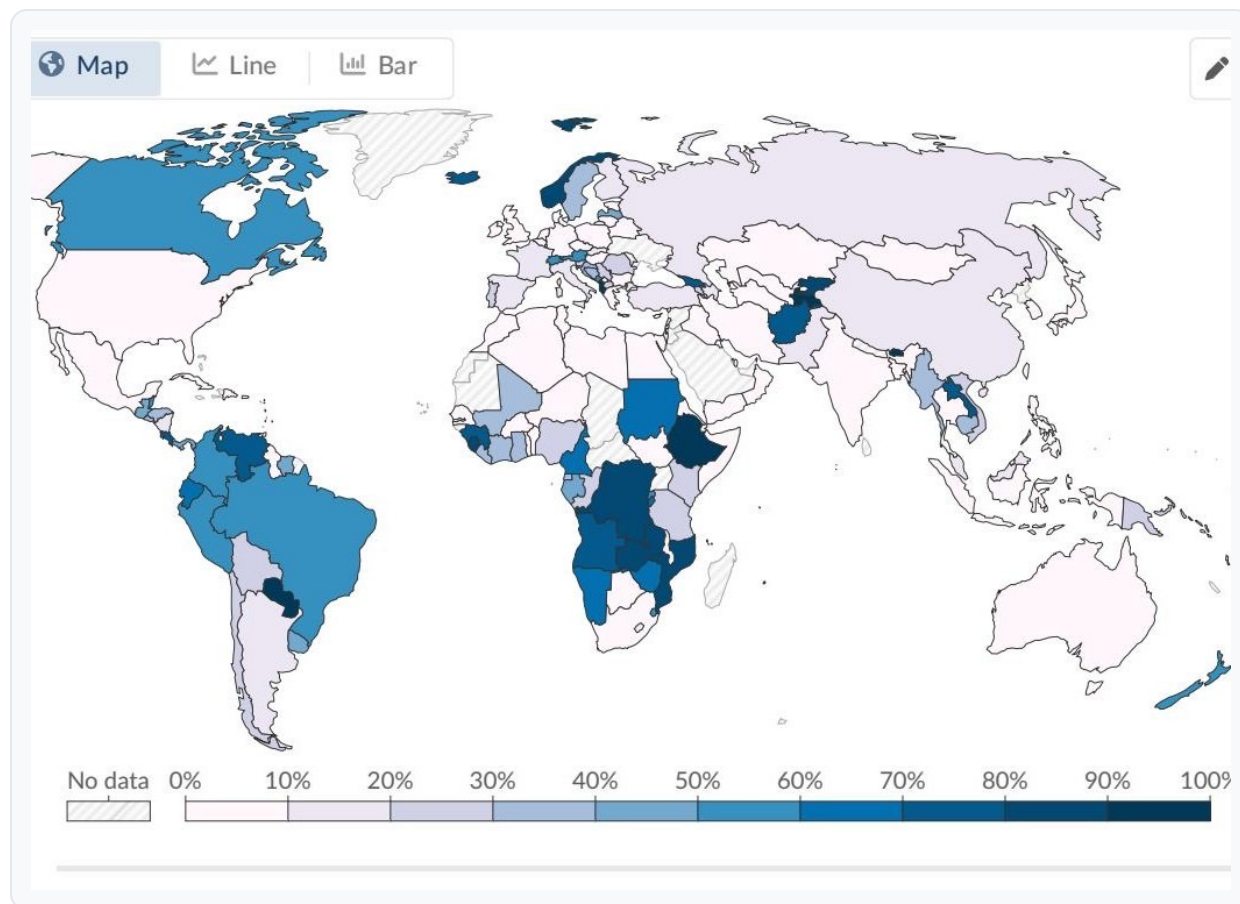


Since 2010

Exponential growth

KEY INSIGHT

The transition to renewables is highly uneven across regions



Regional Leaders (60-90%+ renewable electricity)

Scandinavia

Norway ~98%, Sweden ~65%
Abundant hydropower resources

Latin America

Brazil ~85%, Costa Rica ~99%
Strong hydropower & geothermal

Strong Performers (Wind/Solar Investment)

Denmark & Germany

Heavy investment in wind and solar

Regional Laggards (<10% renewable electricity)

Sub-Saharan Africa

Minimal renewable electricity
infrastructure

Many below 10%

Middle East

Almost entirely dependent
on fossil fuels for electricity

Saudi Arabia <2%

01 // OVERALL PROGRESS

~14% of Primary Energy, ~30% of Electricity

Renewables now supply approximately one-seventh of primary energy and almost one-third of electricity — a significant milestone

02 // HYDROPOWER LEAD

Hydropower Still Dominant at ~50%

Generating roughly half of all renewable electricity, but its growth potential is limited vs. wind and solar

03 // FASTEST GROWTH

Wind + Solar Nearly Match Hydro Output

Wind and solar are the fastest-growing energy technologies in history — combined, nearly as much as hydropower

04 // REGIONAL GAPS

Stark Regional Disparities Persist

Scandinavia and Latin America exceed 60-90% renewable electricity; Middle East and Sub-Saharan Africa below 10%

05 // SECTOR LAG

Transport & Heating Lag Far Behind

Keeping overall primary energy share much lower than electricity share

CALL TO ACTION

The Pace of Transition Must Accelerate

Current growth is insufficient to meet climate targets